Lecture 02b – Sampling designs II

ENVX2001 Applied Statistical Methods

Januar Harianto

Mar 2026

Welcome back

Before the break

We used simple random sampling on the soil carbon data and found:

- SRS mean: **67.3 t/ha**, 95% CI: **(49.87, 84.73)**
- That interval is wide — about ±26% of the mean

We ended with a question: **can we do better with a smarter sampling design?**

What is next

1. Why simple random sampling does not always work well
2. Stratified random sampling — using known groups to sample more efficiently
3. Monitoring — estimating change over time

Simple random sampling works..

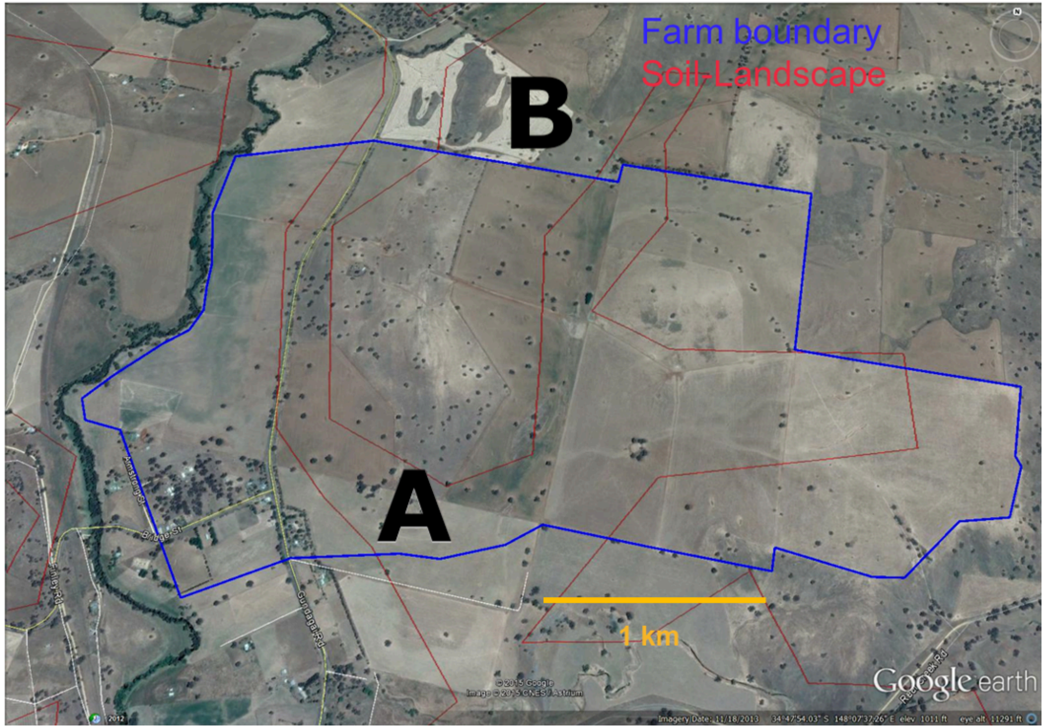
Until it doesn't.

Random sampling on a landscape

```
{ojs}
//| echo: false
import { samplingWidget } from "../../assets/js/sampling-widget.js"
samplingWidget()
```

How do we guarantee that with few samples, we get a good representation of the landscape?

The farmer has extra information



The farmer knows there are two land types on the property:

- Land type A covers 62% of the area, land type B covers 38%
- With simple random sampling, we would expect more samples from type A — but this is not guaranteed

- **Can we use this information to sample more efficiently?**

~~Simple~~ Stratified random sampling

Stratified random sampling

3 steps

1. **Divide** the population into **homogeneous** subgroups (strata).
2. **Sample** from each stratum using simple random sampling.
3. **Pool** (combine) the estimates from each stratum to get an overall population estimate.

Example

If studying plant biodiversity in a national park:

- Divide the park into strata (e.g. forest, grassland, wetland)
- Take random samples within each habitat type
- Combine data to estimate overall biodiversity, weighting by each habitat's area

It’s not as complicated as it sounds...

```
{ojs}
//| echo: false
import { stratifiedStepsWidget } from "../../assets/js/sampling-widget.js"
stratifiedStepsWidget()
```

Strata rules

Strata are...

- **Mutually exclusive and collectively exhaustive** — every unit belongs to exactly one stratum, with no overlaps and nothing left out
- **Homogeneous** — units within a stratum should be more similar to each other than to the population as a whole
- **All sampled** — every stratum must be represented

Which of these are good strata?

The farmer wants to refine their soil carbon estimate. Which stratification would work?

A. Soil type — clay soils vs. sandy soils

B. Soil condition — wet patches, compacted areas, eroded areas

A works. Every sample belongs to exactly one soil type. Carbon storage differs between clay and sandy soils, so the strata are distinct.

B fails. A patch of soil can be wet *and* compacted *and* eroded at the same time. The categories overlap — strata must be mutually exclusive.

Advantages

Stratified random sampling addresses three problems at once:

- **Bias** — every stratum is sampled, so the sample is representative of the population
- **Accuracy** — each stratum is represented by a minimum number of units
- **Insight** — we can compare strata and make inferences about subgroups

Does this make simple random sampling obsolete?

No. With large enough samples, the two methods converge. Simple random sampling is still a good default when strata are unknown or the population is fairly homogeneous.

Calculating stratified estimates

The workflow

Once we have our stratified sample, the calculations follow the same logic as simple random sampling

The workflow

but each step must account for the stratified design.

1. **Pooled mean** ($\bar{y}_s$) — weight each stratum mean by the stratum's share of the population
2. **Pooled standard error** — $SE(\bar{y}_s) = \sqrt{\sum w_i^2 \times \frac{s_i^2}{n_i}}$
3. ***t*-critical value** — based on $df = n - L$ and $\alpha = 0.05$
4. **Confidence interval** — $\bar{y}_s \pm t_{n-L} \times SE(\bar{y}_s)$

> **ℹ Note**
>
> From here we use $\bar{y}$ rather than $\bar{x}$ for the sample mean — this is conventional in sampling design literature and helps distinguish stratified estimates from simple estimates.

Accounting for strata using “weight”

- Each stratum contributes to the overall estimate in proportion to its size in the population.
- Most of the time, we use the stratum’s share of the total area (or population) as the weight.
- The overall population estimate is the sum of the weighted estimates from each stratum.

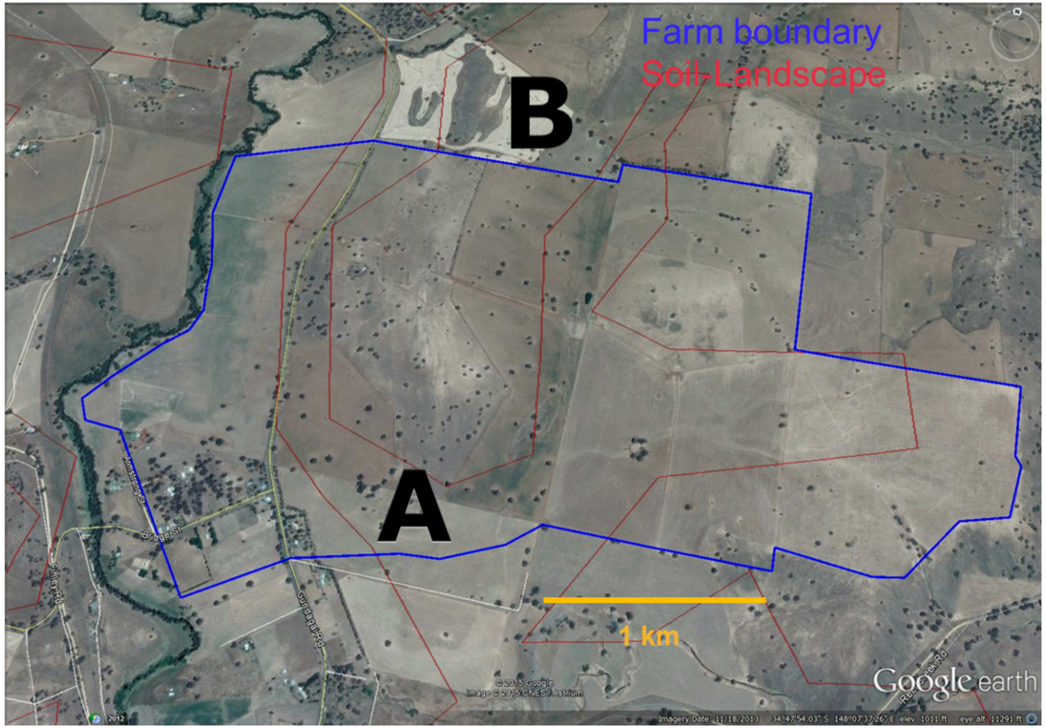
Example

- A forest has two tree types: A (60% of the population) and B (40%).
- We want to estimate the **mean height** of the trees.
- We take 10 height measurements: 7 from type A and 3 from type B.
- The **pooled estimate** for mean height is:

$$0.6 \times \text{mean height of A} + 0.4 \times \text{mean height of B}$$

Soil carbon: setting up the data

Soil carbon content was measured at 7 locations. The amounts were: 48, 56, 90, 78, 86, 71, 42 tonnes per hectare (t/ha).



We know which land type each sample came from:

```r
landA ← c(90, 78, 86, 71)  # stratum A samples (62% of the area)
landB ← c(48, 56, 42)      # stratum B samples (38% of the area)
```

Pooled mean $\bar{y}_s$

> The pooled mean is our best estimate of the overall population mean, taking into account the different stratum sizes.

$$\bar{y}_s = \sum_{i=1}^{L} \bar{y}_i \times w_i$$

We calculate the mean for each stratum ($\bar{y}_i$), multiply by its weight (w_i), and add them together.

Calculating pooled mean: soil carbon example

We first define the weights w_i for each stratum based on their area:

```
weight <- c(0.62, 0.38)  # 62% of area is land type A, 38% is land type B
```

Then we calculate the weighted mean:

```
weighted_mean <- mean(landA) * weight[1] + mean(landB) * weight[2]
weighted_mean
```

```
[1] 68.86833
```

62% of the land has soil carbon like type A, and 38% like type B — the overall estimate reflects both in those proportions.

Pooled standard error of the mean $SE(\bar{y}_s)$

$$SE(\bar{y}_s) = \sqrt{\sum_{i=1}^{L} w_i^2 \times \frac{s_i^2}{n_i}}$$

What is different from simple random sampling?

- Instead of a single variance term, we sum the weighted variances from each stratum
- The w_i^2 term accounts for the relative size of each stratum
- Each stratum contributes its own variance (s_i^2) and sample size (n_i)

t-critical value

Degrees of freedom df

$$df = n - L$$

where n is the total number of samples and L is the number of strata.

- For stratified sampling, we lose one degree of freedom for each stratum mean we estimate
- **Example:** 12 samples across 3 strata gives $df = 12 - 3 = 9$

In R

```r
df <- length(landA) + length(landB) - 2
t_crit <- qt(0.975, df)
t_crit
```

```
[1] 2.570582
```

95 % Confidence interval for stratified random sampling

The formula

$$95\%\ CI = \bar{y}_s \pm t_{n-L}^{0.025} \times SE(\bar{y}_s)$$

where L is the number of strata, n is the total number of samples, and $\bar{y}_s$ is the pooled mean.

- Start with the pooled mean ($\bar{y}_s$)
- Add and subtract the margin of error (t-critical $\times$ pooled SE)
- The result is the range where we are 95% confident the true population mean lies

95 % Confidence interval for stratified random sampling

Putting it all together

```
varA <- var(landA) / length(landA)  # variance of the mean for A
varB <- var(landB) / length(landB)  # variance of the mean for B
weighted_var <- weight[1]^2 * varA + weight[2]^2 * varB
weighted_se <- sqrt(weighted_var)
ci <- c(
  L95 = weighted_mean - t_crit * weighted_se,
  u95 = weighted_mean + t_crit * weighted_se
)
ci
```

```
     L95      u95
61.04864 76.68803
```

The payoff

Simple random vs. stratified random sampling

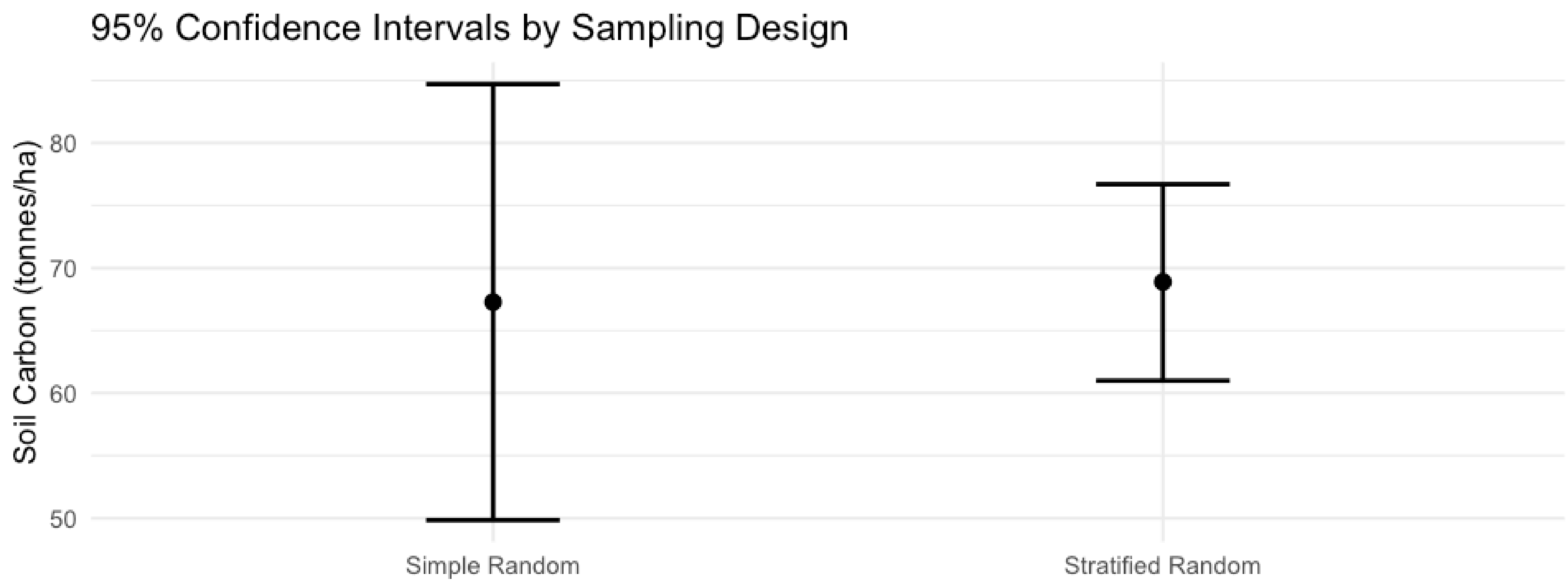
We applied stratified random sampling to the same soil carbon data from the first half. How do the results compare?

```r
library(tidyverse)
# Manually printing the results below as SRS data is in previous lecture
compare <- tibble(
  Design = c("Simple Random", "Stratified Random"),
  Mean = c(67.29, 68.9),
  `Var (mean)` = c(50.83, 9.30),
  L95 = c(49.85, 61),
  U95 = c(84.73, 76.7),
  df = c(6, 5))
knitr::kable(compare)
```

Design	Mean	Var (mean)	L95	U95	df
Simple Random	67.29	50.83	49.85	84.73	6
Stratified Random	68.90	9.30	61.00	76.70	5

Visual comparison of 95% confidence intervals

```r
ggplot(compare, aes(x = Design, y = Mean)) +
  geom_point(size = 3) +
  geom_errorbar(aes(ymin = L95, ymax = U95), width = 0.2, linewidth = 1) +
  labs(title = "95% Confidence Intervals by Sampling Design",
       y = "Soil Carbon (tonnes/ha)",
       x = "") +
  theme_minimal(base_size = 14)
```



Key differences

- Both methods give similar estimates of the mean (~67–69 t/ha)
- Stratified sampling produces a much narrower confidence interval

- The variance of the mean is about 5 times smaller with stratified sampling

Efficiency

How much more precise is stratified sampling?

$$\text{Efficiency} = \frac{\text{Variance of SRS}}{\text{Variance of Stratified}}$$

- Efficiency > 1 means stratified sampling is more precise for the same sample size
- An efficiency of 5 means you would need 5 times as many SRS samples to match the stratified precision

In R

```r
efficiency <- 50.83 / 9.30
efficiency
```

```
[1] 5.465591
```

How many SRS samples would we need for the same precision?

```r
round(7 * efficiency, 0)
```

```
[1] 38
```

About 38 samples with SRS to match what 7 stratified samples achieved.

Tips on implementation

- The hardest part is identifying good strata and assigning units to them
- Common stratification variables in environmental science:
  - **Spatial**: elevation bands, soil types, vegetation zones
  - **Temporal**: seasons, time of day, growth stages
  - **Management**: treatment types, land-use history
- **Sample allocation**: distribute samples across strata in proportion to:
  - the size of the strata (e.g. 60% of area = 60% of samples)
  - the variance within strata (more samples where variation is higher)

The farmer comes back

A year later...

The farmer now has a good estimate of soil carbon on the property. But a year has passed, and new management practices have been introduced.

New question: has soil carbon changed?

To answer this, the farmer measures the same property again. This is a **monitoring study** — we are estimating *change* rather than a single value.

Same sites or new sites?

- Key decision: **how do we select sites for the second measurement?**
  1. Return to the **same sites**?
  2. Select completely **new sites**?
- This choice affects how we analyse the data

Change in mean $\Delta \bar{y}$

- The difference between the means of the two sets of measurements.

$$\Delta \bar{y} = \bar{y}_2 - \bar{y}_1$$

where $\bar{y}_2$ and $\bar{y}_1$ are the means of the second and first set of measurements, respectively.

How precise is our estimate of change?

Variance of the change in mean $Var(\Delta\bar{y})$

$$Var(\Delta\bar{y}) = Var(\bar{y}_2) + Var(\bar{y}_1) - 2 \times Cov(\bar{y}_2, \bar{y}_1)$$

The uncertainty comes from both measurements. But if we sample the same sites twice, the **covariance** term usually *reduces* the overall uncertainty.

What is covariance? Consider three of the farmer's sites:

- Site 1: First visit = 90 t/ha, Second visit = 95 t/ha
- Site 2: First visit = 48 t/ha, Second visit = 52 t/ha
- Site 3: First visit = 71 t/ha, Second visit = 75 t/ha

Sites with high carbon in the first measurement still have high carbon in the second. This positive relationship is covariance — and it reduces uncertainty in the estimate of change.

Paired vs independent sampling

Quick decision guide

1. **Same sites?** Use paired approach:
   - Sites are the same in both visits
   - Use paired t-test
   - Account for covariance between visits
2. **Different sites?** Use independent approach:
   - New random sites in second visit
   - Use two-sample t-test
   - No covariance between visits

Returning to the same sites (paired sampling) usually gives more precise estimates because it removes site-to-site variation.

Confidence interval for change

$$95\%\ CI = \Delta\bar{y} \pm t^{0.025}_{n-1} \times SE(\Delta\bar{y})$$

The standard error of the change, $SE(\Delta\bar{y})$, accounts for the relationship between the two measurements when the same sites are visited. This is not straightforward to calculate by hand.

In practice, R handles these calculations using `t.test()`:

- For same sites: `paired = TRUE`
- For different sites: `paired = FALSE`

R connection

`t.test()` with `paired = TRUE` performs a paired t-test — you will use this in Lab 02 to analyse monitoring data.

Thanks!

Questions?